

Bugs found: 3

1. The 'BadKey' class overrides 'equals' but fails to override 'hashCode', violating the contract for use in a HashMap.
2. Array index out of bounds error occurs when attempting to access 'objs[5]' on an array of size 2.
3. A NullPointerException is triggered by calling the 'length()' method on a null reference 'n'.